

Software Architecture Document (SAD)

NexGenius Solutions Website

Version: 1.0

Document Type: Software Architecture Document (SAD)

Project: NexGenius Solutions Website

Architecture Style: Modern Jamstack + Serverless Architecture

1. Introduction

Purpose

This document defines the overall software architecture for the NexGenius Solutions website. It describes the system structure, technology choices, component interactions, deployment architecture, and future scalability considerations.

The architecture is designed to be modern, secure, SEO-friendly, highly performant, and easy to maintain while supporting future business growth.

2. Architecture Goals

The system should:

- Deliver exceptional performance worldwide
 - Rank well in search engines
 - Capture and manage business leads
 - Support Pakistani and international clients
 - Be scalable for future AI products
 - Be easy to maintain and extend
 - Minimize operational costs
 - Provide enterprise-grade security
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3. Architecture Style

Primary Architecture

Modern Serverless Architecture

Browser



Next.js Application



Server Actions / API Routes



Third-Party Services



Company Email

Design Principles

- Component-Based Architecture
 - Server Components First
 - Client Components Only When Needed
 - Modular Feature-Based Structure
 - Separation of Concerns
 - Reusable UI Components
 - Type Safety
 - Performance First
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4. High-Level System Architecture

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User Browser

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Next.js Frontend

(Server Components)

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API Routes Static Content

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Email Service MDX Blog

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Company Inbox

External Integrations

- WhatsApp
- Google Analytics
- Google Search Console
- Microsoft Clarity

5. Technology Decisions

Frontend

Framework

Next.js 15

Reason

- Excellent SEO
 - Server Rendering
 - Static Generation
 - Image Optimization
 - Performance
 - Easy Deployment
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React 19

Reason

Component Architecture

TypeScript

Reason

- Type Safety
 - Better DX
 - Maintainability
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Tailwind CSS

Reason

- Utility First
 - Small Bundle
 - Easy Customization
-

shadcn/ui

Reason

- Accessible Components
 - Customizable
 - Modern Design
-

Framer Motion

Reason

Premium Animations

6. Backend Architecture

Current Phase

No dedicated backend server.

Business logic handled through:

- API Routes
- Server Actions

Advantages

- Lower Cost
- Simpler Deployment
- Faster Development
- Less Infrastructure

Future Upgrade Path

Next.js

↓

NestJS API

↓

PostgreSQL

↓

Redis



Queue System



AI Services

7. Component Architecture

Core Components

Navigation

Hero

Services

Technology Stack

About

Blog

Contact

Footer

Reusable Components

Button

Card

Badge

Modal

Input

Textarea

Accordion

FAQ

CTA

Each component should:

- Be reusable
- Be typed
- Be independently testable
- Support dark mode (future)

8. Routing Architecture

/

Home

/about

/services

/blog

/blog/[slug]

/contact

404

500

API

/api/contact

/api/revalidate (future)

/api/newsletter (future)

9. Content Architecture

Current

MDX

Benefits

- Fast
- Git Versioned
- SEO Friendly
- No CMS Cost

Future

Headless CMS

Options

- Sanity
 - Contentful
 - Strapi
-

10. Data Flow

Contact Form

User

↓

Frontend Validation

↓

API Route

↓

Server Validation

↓

Email Provider



Company Email



Success Response

Blog Page

User



Next.js



MDX



Rendered HTML



Browser

11. External Integrations

Communication

- WhatsApp

Email

- Resend
- Brevo

Analytics

- Google Analytics
- Search Console
- Microsoft Clarity

Future

- Calendly
- HubSpot
- Stripe
- PayPal
- CRM

12. Security Architecture

Input Validation

Required

Rate Limiting

Required

Spam Protection

Google reCAPTCHA

Environment Variables

Required

HTTPS

Required

Email Validation

Required

Future

- Authentication
- RBAC

- Audit Logs
-

13. Performance Strategy

Server Components

Image Optimization

Code Splitting

Lazy Loading

Font Optimization

Static Rendering

Caching

CDN Delivery

Target

Lighthouse

95+

14. SEO Architecture

Metadata API

Structured Data

Sitemap

Robots

Canonical URLs

Open Graph

Twitter Cards

Schema.org

Dynamic Metadata

Blog SEO

15. Scalability Strategy

Phase 1

Static Website

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Phase 2

CMS

↓

Phase 3

Authentication

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CRM

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Client Dashboard

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AI Platform

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Multi-Tenant SaaS

The architecture should support gradual expansion without requiring a complete rebuild.

16. Deployment Architecture

Developer

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GitHub

↓

Vercel

↓

Production

Automatic deployment on every push to the main branch.

Environment variables managed through Vercel.

17. Monitoring

Google Analytics

Microsoft Clarity

Vercel Analytics (Optional)

Error Monitoring (Future)

- Sentry
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18. Coding Standards

TypeScript Strict Mode

ESLint

Prettier

Component Naming

PascalCase

Hooks

camelCase

Folder Names

kebab-case

19. Future Architecture Vision

NexGenius Solutions Website

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Lead Generation Platform

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CRM

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Client Portal

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Project Management

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AI Dashboard

↓

Knowledge Base

↓

Subscription Billing

↓

20. Architectural Decisions (ADR)

Decision	Choice	Reason
Frontend	Next.js 15	SEO + Performance
Language	TypeScript	Maintainability
Styling	Tailwind CSS	Fast UI Development
UI Library	shadcn/ui	Modern Components
Backend	API Routes	Simple MVP
Email	Resend	Reliable Delivery
Hosting	Vercel	Easy CI/CD
Blog	MDX	Cost Effective
Analytics	GA4 + Clarity	User Insights
Deployment	GitHub + Vercel	Automated Workflow

21. Risks & Mitigations

Risk: Spam submissions
Mitigation: reCAPTCHA + Rate Limiting

Risk: Email delivery failures
Mitigation: Retry logic and provider monitoring

Risk: Performance degradation
Mitigation: Static rendering, image optimization, lazy loading

Risk: Future feature growth
Mitigation: Modular architecture with clear separation of concerns

22. Conclusion

The NexGenius Solutions architecture follows a modern, serverless-first approach that balances performance, SEO, maintainability, and scalability. It minimizes operational complexity for the MVP while

providing a clear migration path toward a full-featured AI platform with authentication, CRM capabilities, client dashboards, and enterprise integrations as the business grows.